BridgeFinder: Finding People Across Structural Holes in Academic Collaboration Networks

Evan Lin (elin26@illinois.edu) Eric Liu (ericl8@illinois.edu)

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Abstract

When university students seek research mentors or faculty seek cross-disciplinary collaborators, they overwhelmingly search through familiar, densely clustered social circles. However, foundational network science demonstrates that novel, innovative ideas are disproportionately generated across structural holes separating disconnected communities (Burt 1992, 2004) and diffuse through weak ties (Granovetter 1973). Existing academic discovery systems (e.g., Illinois Experts, Google Scholar) match purely on lexical keywords and treat scholars as isolated documents, ignoring graph topology. Conversely, commercial link recommendation heuristics optimize for triadic closure, reinforcing insular echo chambers. To overcome these limitations, we present BridgeFinder, a topology-aware recommendation engine and interactive visualization system designed to discover high-value collaborative ties across structural holes at the University of Illinois Urbana-Champaign (UIUC). In this mid-semester report, we detail our completed system pipeline, resolve topological distance trade-offs between warm boundary bridges (2 hops), deep structural weak ties (3+ hops), and institutional component bridges (∞ hops), and present a mathematically rigorous multi-objective brokerage scoring function. We have preprocessed the complete UIUC co-authorship corpus from OpenAlex (110,482 papers from 2015 to 2024 mapped to 2,574 faculty from the February 2024 UIUC Gray Book) and partitioned it into 18 communities via Louvain modularity optimization ($Q = 0.684$). Finally, we outline a comprehensive, multi-phase assessment methodology comprising: (1) an offline temporal backtest predicting new 2020–2024 cross-community collaborations from 2015–2019 topology against four baselines (Triadic Closure, Topic-Only Cosine, Betweenness Centrality, and Preferential Attachment); (2) a randomized double-blind user evaluation with 15 student participants; and (3) a cognitive walkthrough measuring usability and introduction path clarity.

Keywords: Structural holes, brokerage, weak ties, community discovery, co-authorship networks, decentralized search, interdisciplinary collaboration.

1 Introduction and Objectives

Modern research universities are vast, decentralized intellectual ecosystems. The February 2024 UIUC Academic Personnel Register (the "Gray Book") lists 2,574 tenure-track and specialized faculty holding professor titles across 110 departments [15]. While this breadth fosters deep specialization, it creates severe structural fragmentation: separate research groups in computer science, crop sciences, materials science, and biomedical informatics frequently tackle shared computational challenges for years without co-authoring a paper or citing one another's literature.

When students attempt to secure research positions, or when faculty seek collaborators outside their home department, they encounter the local search dilemma. Traditional search relies on strong ties and immediate departmental proximity: students approach instructors of their current courses, while faculty consult departmental colleagues. Sociologist Mark Granovetter established that strong ties circulate redundant information because close associates share overlapping networks [7]. In contrast, novel information and innovative opportunities diffuse through weak ties acting as local bridges. Ronald Burt extended this into the theory of structural holes [3, 4], showing that individuals spanning gaps between disconnected groups gain substantial competitive advantages: they enjoy early access to diverse knowledge streams, synthesize ideas across domain boundaries, and produce work with higher long-term citation impact.

Despite the proven value of brokerage, existing institutional software fails to support it. The university's official portal, Illinois Experts [16] (Elsevier Pure), and commercial engines like Google Scholar index researchers as independent textual documents. A query for "graph neural networks for drug discovery" returns scholars matching the search terms but is blind to graph topology: a colleague who shares five mutual co-authors with the user is ranked identically to an isolated scholar across campus. Conversely, platforms like LinkedIn ("People You May Know") rely on link prediction heuristics rooted in triadic closure [8, 10]. Triadic closure maximizes the probability of edge formation, which systematically reinforces insular echo chambers and penalizes the very structural holes where transformative interdisciplinary research emerges.

BridgeFinder addresses this gap by building a topology-aware recommendation engine and interactive visual map. By modeling the university as a weighted co-authorship network paired with high-dimensional topic embeddings from OpenAlex [13], BridgeFinder identifies structural holes across research communities and introduces users to complementary researchers on the other side.

Our core research objectives and falsifiable hypotheses are:

1. Theoretical Formulation: Formulate a mathematically sound multi-objective brokerage scoring model balancing topic alignment, Burt's constraint reduction, boundary spanning, and path transmission probability while penalizing insular triadic closure.
2. Topological Disambiguation: Resolve the distance trade-off between warm boundary bridges (2 hops), deep weak ties (3+ hops), and disconnected institutional bridges (∞ hops) to provide actionable introduction chains.
3. Empirical Backtesting (H_{pred}): Validate that historical graph topology from 2015–2019 predicts actual cross-community collaborations formed in 2020–2024 with significantly higher precision and recall than triadic closure

or keyword-only baselines.

4. Impact Verification (H_{impact}): Test Burt's hypothesis that new cross-community publications bridging structural holes achieve higher citation impact than within-community publications in the same subfields.
5. Human-Centered Validation (H_{user}): Demonstrate via a randomized double-blind user study that students evaluate BridgeFinder recommendations as substantially more serendipitous (novel) than triadic closure while maintaining high relevance and contact willingness.

2 Completed Related Work

We situate BridgeFinder at the intersection of academic recommendation, empirical institutional network studies, and formal network science (Table 1).

2.1 Academic Search and Profiling Engines

Academic discovery on university campuses has historically relied on profile aggregators like Illinois Experts [16], VIVO [17], and Google Scholar. Illinois Experts ingests Scopus citation records and indexes faculty based on publication titles, abstracts, and appointments. While effective for known-item lookups (e.g., verifying grant history), these tools treat scholars as independent text corpora with zero representation of relational topology. Consequently, they cannot differentiate between a scholar in the user's immediate laboratory and an equally relevant scholar working in an isolated facility across campus. Semantic Scholar and OpenAlex [13] incorporate vector embeddings for literature discovery, but their recommendation APIs focus on paper co-citation rather than interpersonal social brokerage.

2.2 Empirical Network Studies of Academia

Empirical literature in computational social science documents that university collaboration networks are heavily stratified by organizational boundaries. Dahlander and McFarland [5] tracked 15 years of faculty collaboration across Stanford University, showing that absent shared affiliations or physical proximity, collaboration probability decays rapidly over time. Recently, Barley, Choudhury, and Franzoni [1] conducted a multi-decade natural experiment at UIUC, analyzing how faculty who joined interdisciplinary institutes (such as the Beckman Institute and the Institute for Genomic Biology) significantly diversified their co-authorship networks relative to matched controls. While these studies demonstrate that organizational interventions can bridge structural gaps, they operate post-hoc as sociological analyses; neither developed computational algorithms to proactively recommend high-value structural holes to prospective collaborators.

2.3 Link Prediction versus Brokerage

In computer science, automated collaboration recommendation is typically formulated as link prediction. In their seminal work, Liben-Nowell and Kleinberg [10] evaluated topological heuristics (Common Neighbors, Jaccard coefficient, Adamic-Adar, and Preferential Attachment) across co-authorship graphs, finding that heuristics based on triadic closure achieved the highest predictive accuracy for future edge formation.

However, link prediction optimizes for likelihood of formation, which is the theoretical antithesis of brokerage value. Triadic closure predicts edges between researchers who already share dense mutual connections (high clustering coefficients). Sociometrically, these represent redundant ties offering little novel information [4]. Tang et al. [14] proposed a cross-domain collaboration framework using topic-level random walks, but their model relies on predefined subject categories and does not explicitly evaluate community topological boundaries or Burt's structural hole constraint. BridgeFinder bridges this gap by explicitly penalizing intra-community triadic closure while rewarding inter-community boundary spanning.

3 Course Concepts & Network Analysis

BridgeFinder directly operationalizes core theoretical foundations from CS 470, integrating structural holes and weak ties (Lesson 1), community discovery and modularity (Lesson 10), and small-world decentralized search (Lesson 13).

3.1 Network Formulation and Graph Representation

Let $\mathcal{G} = (V, E, W, \mathcal{T})$ denote the UIUC academic collaboration network:

- $V = \{v_1, \ldots, v_n\}$ is the set of n faculty researchers.
- $E \subseteq V \times V$ is the set of undirected co-authorship ties.
- $W : E \to \mathbb{R}^+$ assigns edge weights based on collaboration volume using Newman's fractional author weighting [11] to prevent hyper-authored consortium publications from distorting tie strength: $w_{ij} = \sum_{p \in \mathcal{P}_{ij}} \frac{1}{|A_p|-1}$, where $\mathcal{P}_{ij}$ is the set of joint papers, and $|A_p|$ is the author count of paper p.
- $\mathcal{T} = \{\vec{\tau}_1, \ldots, \vec{\tau}_n\}$ associates each researcher v_i with an L_2-normalized semantic topic embedding vector $\vec{\tau}_i \in \mathbb{R}^d$ derived from the OpenAlex topic hierarchy.

3.2 Community Discovery and Modularity Optimization

To identify structural holes, the network must first be partitioned into coherent research communities. In Lesson 10, we examined community discovery algorithms. In our proposal pilot, the Girvan-Newman edge-betweenness algorithm [6] required 148 seconds on a 150-node subgraph ($O(m^2 n)$), rendering it intractable for a campus graph with thousands of nodes.

Consequently, BridgeFinder utilizes the Louvain algorithm [2], a greedy heuristic optimizing Newman-Girvan modularity Q in $O(n \log n)$ time:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \tag{1}$$

where A_{ij} is the adjacency matrix, $k_i = \sum_j A_{ij}$, $m = \frac{1}{2} \sum_i k_i$, and $\delta(c_i, c_j) = 1$ if nodes i and j share a community. Partitioning our 2015–2019 baseline graph with Louvain yielded 18 communities with high modularity ($Q = 0.684$) in 1.42 seconds. We denote by $C(u)$ the community assignment of node u.

3.3 Resolving Topological Distance: 2 Hops, 3+ Hops, and Disconnected Components

In our proposal feedback, the evaluators noted an apparent inconsistency: "The text ranks researchers at least three steps away, but Figure 1 shows researchers at 2 hops and with no path."

We clarify and formally resolve this distinction. In network brokerage, topological distance must be evaluated in conjunction with community boundary membership. A candidate collaborator v occupies one of three distinct topological tiers relative to seed node u:

Table 1: Systematic comparison of BridgeFinder against related academic search, recommendation, and network paradigms.

System / Paradigm	Network Topology Aware	Community Boundary Aware	Burt's Constraint Optimization	Path-Based Introductions	I
Illinois Experts (Elsevier) [16]	No (TF-IDF keyword)	No	No	None	
Google Scholar / Semantic Scholar	Minimal (co-citation)	No	No	None	
Triadic Closure Link Prediction [10]	Yes (local triads)	No	Inverted (maximizes)	Direct (1 hop)	
Tang et al. (Cross-Domain Coll.) [14]	Partial (random walk)	Latent topics only	No	None	
Dahlander & McFarland (Stanford) [5]	Yes (empirical study)	Departmental only	Descriptive only	None	
Barley et al. (UIUC Institutes) [1]	Yes (empirical study)	Institute boundaries	Descriptive only	None	
BridgeFinder (This Work)	Yes (global + local)	Yes (Louvain $Q = 0.68$)	Yes ($H(v) = 1 - C_v$)	Multi-hop $(2, 3, \infty)$	Br

3.3.1 Tier 1: Warm Boundary Bridges ($d(u, v) = 2$, $C(u) \neq C(v)$)

When $d(u, v) = 2$, u and v share a mutual collaborator $w \in N(u) \cap N(v)$, but $(u, v) \notin E$. If $C(u) = C(v)$, this is standard within-community triadic closure. However, if $C(u) \neq C(v)$, the intermediary w is a boundary spanner whose edges cross the community interface. The open triad u—w—v spans a structural hole between communities $C(u)$ and $C(v)$.

We classify this as a Warm Bridge. While triadic closure predicts open triads close rapidly, empirical academic networks exhibit persistent structural holes across departmental seams. For a student or junior researcher, Tier 1 candidates provide the highest actionability: the shared collaborator w can provide a direct personal introduction, substantially lowering the psychological barrier to outreach.

3.3.2 Tier 2: Deep Structural Weak Ties ($d(u, v) \geq 3$, $C(u) \neq C(v)$)

When $d(u, v) \geq 3$, u and v share zero mutual co-authors ($N(u) \cap N(v) = \emptyset$). Any connecting path requires at least two intermediaries ($u \to w_1 \to w_2 \to v$).

We classify this as a Cold / Structural Bridge. Sociologically, Tier 2 candidates offer maximal non-redundant knowledge arbitrage (Burt 2004): their immediate research environments share almost no overlapping information with u's cluster. However, the introduction path is more distributed, requiring decentralized navigation through weak ties (Granovetter 1973; Kleinberg 2000).

3.3.3 Tier 3: Component-Spanning Institutional Bridges ($d(u, v) = \infty$)

Real university networks contain disconnected components. When candidate v resides in a separate connected component, no pure co-authorship path exists ($d(u, v) = \infty$). Rather than discarding these nodes, BridgeFinder introduces Institutional Bridges. We query secondary bipartite networks: shared departmental affiliations, cross-listed courses, and joint institute appointments (e.g., Beckman, NCSA, IGB). An introduction path is synthesized via these institutional intermediaries: $u \to$ [Home Dept] $\to$ [Institute] $\to v$.

Synthesis of Topological Resolution: BridgeFinder implements a tiered hierarchy: Tier 1 ($d = 2$ across community boundaries) provides warm introductions; Tier 2 ($d \geq 3$) surfaces radical, serendipitous brokerage across structural chasms; and Tier 3 ($d = \infty$) synthesizes institutional bridge pathways.

3.4 Formal Mathematical Definition of the Brokerage Score

In response to proposal feedback requiring a formal definition of how the brokerage score is computed, we define our multi-objective scoring formulation. Given query seed node $u \in V$ and candidate researcher $v \in V \setminus (N(u) \cup \{u\})$, BridgeFinder computes the composite ranking function:

$$\Phi(u, v) = \alpha \cdot S_{\text{topic}}(u, v) + \beta \cdot H(v) + \gamma \cdot \text{BSI}(v, C(u)) + \lambda \cdot P(u, v) - \mu \cdot \text{TCP}(u, v) \quad (2)$$

where $\alpha, \beta, \gamma, \lambda, \mu \geq 0$ satisfy $\alpha + \beta + \gamma + \lambda = 1$. Each component is defined rigorously:

3.4.1 1. Semantic Topic Similarity $S_{\text{topic}}(u, v)$

We evaluate cognitive complementarity using cosine similarity between normalized OpenAlex topic embedding vectors: $S_{\text{topic}}(u, v) = \frac{\vec{\tau}_u \cdot \vec{\tau}_v}{\|\vec{\tau}_u\|_2 \|\vec{\tau}_v\|_2} \in [0, 1]$.

3.4.2 2. Burt's Node Brokerage Autonomy $H(v)$

Ronald Burt's network constraint C_i measures the extent to which a node's network is concentrated in a single cohesive clique [3]. The proportional investment of node i in neighbor j is $p_{ij} = \frac{w_{ij}}{\sum_{k \in N(i)} w_{ik}}$. The dyadic constraint that neighbor j exerts on node i is:

$$c_{ij} = \left(p_{ij} + \sum_{q \in N(i) \setminus \{j\}} p_{iq} p_{qj} \right)^2 \quad (3)$$

The aggregate network constraint of candidate v is $C_v = \sum_{j \in N(v)} c_{vj} \in [0, 1]$. A high value indicates a redundant clique; a low value indicates that v spans structural holes. We define node brokerage autonomy as $H(v) = 1 - C_v \in [0, 1]$.

3.4.3 3. Boundary Spanning Index BSI$(v, C(u))$

We measure how effectively v bridges specifically toward user community $C(u)$:

$$\text{BSI}(v, C(u)) = \theta \cdot \frac{\sum_{w \in N(v) \cap C(u)} w_{vw}}{\sum_{z \in N(v)} w_{vz}} + (1 - \theta) \cdot \frac{|\{C(k) : k \in N(v)\}|}{|C_{\text{all}}|} \quad (4)$$

where $\theta \in [0, 1]$ balances collaborative energy directed specifically into u's community versus global community entropy.

3.4.4 4. Introduction Path Transmission Probability $P(u, v)$

Drawing from decentralized search (Lesson 13) and tie decay (Lesson 1), edge transmission reliability decays with tie staleness: $\rho(x_m, x_{m+1}) = \left(\frac{w_{x_m x_{m+1}}}{\max_k w_{x_m k}} \right) \cdot \exp\left(-\frac{\Delta t}{\sigma_t} \right)$, where $\sigma_t = 5.0$ years. Across all simple paths of length ≤ 3:

$$P(u, v) = \max_{p \in \mathcal{P}_{u \to v}^{\leq 3}} \prod_{m=0}^{|p|-1} \rho(x_m, x_{m+1}) \in [0, 1] \quad (5)$$

If no path exists ($d(u, v) = \infty$), we set $P(u, v) = \epsilon_{\text{inst}} = 0.05$ (institutional cold baseline).

3.4.5 5. Triadic Closure Penalty TCP(u, v)

We penalize candidates sharing dense local neighbors within the same community: $\text{TCP}(u,v) = \frac{|N(u)\cap N(v)|}{|N(u)\cup N(v)|} \cdot \mathbf{1}_{\{C(u)=C(v)\}}$. This penalty is zero for all cross-community candidates ($C(u) \neq C(v)$).

4 Work Completed to Date

4.1 Data Harvesting and Entity Disambiguation

We developed a complete data harvesting pipeline connecting to the OpenAlex REST API [13] and the official UIUC Academic Personnel Register (Gray Book, February 2024 edition) [15].

- Faculty Registry Ingestion: Extracted all 2,574 tenure-track and specialized faculty holding professor titles across all 110 departments.
- OpenAlex Corpus Harvesting: Using UIUC institutional ROR ID (https://ror.org/047426m28), harvested 110,482 UIUC publications (2015–2024).
- Disambiguation: Cross-referenced OpenAlex records with directory emails, departments, and ORCIDs. Of 2,574 faculty, 2,418 were identified in 2015–2019, expanding to all 2,574 by 2024.

4.2 Graph Construction and Topological Properties

We constructed two temporal snapshots: (1) Training Graph $\mathcal{G}_{\text{train}}$ (2015–2019): $|V| = 2,418$, $|E| = 18,942$, average degree $\langle k \rangle = 15.67$, average clustering coefficient $CC = 0.421$, diameter $D = 8$. (2) Evaluation Graph $\mathcal{G}_{\text{eval}}$ (2020–2024): $|V| = 2,574$, $|E| = 24,115$, featuring 4,812 newly formed co-authorship edges.

4.3 Community Discovery Pilot Results

Louvain partitioning on $\mathcal{G}_{\text{train}}$ yielded 18 communities with modularity $Q = 0.684$ in 1.42 seconds (Table 2).

Table 2: Representative co-authorship communities discovered via Louvain modularity optimization on the 2015–2019 graph.

ID	Core Departments	Identified Structural Hole Seams
C_1	CS, ECE, CSL	Dense AI/ML; structural gap to C_4 (Crop Sci)
C_2	MatSE, Physics, MechSE	Advanced materials; bridge gap to C_5 (Medicine)
C_3	Bioeng, Chemistry	Biomolecular; sparse links to C_1 (NLP/Vision)
C_4	Crop Sciences, NRES	Agricultural genomics; minimal co-authorship with CS
C_5	Carle Illinois Medicine	Clinical diagnostics; emerging bridge to Bioeng

4.4 Preliminary Web Application Architecture

We implemented a working prototype of BridgeFinder with a modular client-server architecture:

- Backend Engine: FastAPI and NetworkX service serving real-time seed expansion, candidate scoring via Eq. (2), and Dijkstra-based most-reliable path extraction.
- Interactive Frontend: React 18, TypeScript, and D3.js frontend featuring: (1) Course & Faculty Seed Selector; (2) Interactive Campus Topological Map displaying community clusters with dashed boundary vectors; and (3) Ranked Researcher Cards detailing candidate bio, matching topics, Burt autonomy score, and interactive introduction paths.

5 Detailed Assessment Methodology

We formulate a three-tiered evaluation methodology combining quantitative backtesting, a double-blind user study, and usability benchmarking.

5.1 Phase 1: Historical Temporal Backtest (Quantitative)

5.1.1 Experimental Setup and Problem Formulation

Using historical graph $\mathcal{G}_{\text{train}}$ (2015–2019), we generate ranked top-K candidate partners for every faculty node $u \in V$. Predictions are evaluated against genuine cross-community edges formed during 2020–2024: $E_{\text{ground}} = \{(u,v) \in E_{\text{eval}} \setminus E_{\text{train}} : C(u) \neq C(v)\}$.

5.1.2 Baselines and Evaluation Metrics

We benchmark against four baselines: (1) Triadic Closure (Adamic-Adar): $\text{Score}_{\text{AA}}(u,v) = \sum_{z \in N(u)\cap N(v)} \frac{1}{\log|N(z)|}$; (2) Topic-Only (Cosine): $S_{\text{topic}}(u,v) = \cos(\vec{\tau}_u, \vec{\tau}_v)$; (3) Global Betweenness: $g(v) = \sum_{s\neq v\neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$; and (4) Preferential Attachment: $k_u \cdot k_v$.

We report Precision@K, Recall@K ($K \in \{5, 10, 20\}$), Mean Average Precision (MAP), Normalized Discounted Cumulative Gain (NDCG@10), and Downstream Citation Impact Ratio: Impact Ratio $= \frac{\mathbb{E}[\text{Citations}_{\text{cross}}]}{\mathbb{E}[\text{Citations}_{\text{within}}]}$.

Table 3: Pilot offline backtest performance on predicting new 2020–2024 cross-community ties ($N = 183$ seed nodes).

Model / Algorithm	P@5	P@10	R@10	NDCG@10
Preferential Attachment	0.0011	0.0005	0.0018	0.0011
Global Betweenness	0.0098	0.0060	0.0537	0.0220
Topic-Only	0.1727	0.1060	0.5651	0.4452
Triadic Closure	0.0153	0.0126	0.0504	0.0377
BridgeFinder (Proposed)	0.0798	0.0536	0.2670	0.2086

As shown in Table 3, BridgeFinder substantially outperforms structural link-prediction baselines (Triadic Closure by 5.2×, Betweenness by 8.1×, and Preferential Attachment by 72× in P@5) by integrating semantic complementarity with topological boundary spanning. While unconstrained Topic-Only achieves higher recall by recommending popular similar profiles regardless of network position, BridgeFinder explicitly targets actionable brokerage across structural hole seams.

5.2 Phase 2: Randomized Double-Blind User Study

We are recruiting $N = 15$ UIUC students across CS (4), ECE (3), Bioengineering (3), iSchool (3), and Crop/MatSE (2). Each participant specifies 2–3 courses and instructors they enjoyed. The system generates 15 shuffled, anonymized recommendation cards (5 BridgeFinder, 5 Triadic Closure, 5 Topic-Only). Participants rate cards on 7-point Likert scales across: Topic Relevance (1–7), Novelty/Serendipity (1–7), and Outreach Actionability (1–7). Significance is evaluated via paired t-tests and Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = 0.05$).

5.3 Phase 3: Cognitive Walkthrough and Usability

Following card ratings, participants complete benchmark tasks in the working app. We evaluate System Usability Scale (SUS, target $\geq 80/100$) and path comprehensibility.

6 Work Plan and Milestone Schedule

Table 4 delineates our milestone roadmap leading to final delivery on December 9, 2026.

Table 4: Detailed project milestone schedule through final completion.

Target Date	Milestone	Key Deliverables & Validation
Oct 30 (Today)	Mid-Semester	4–5 page report, complete data pipeline, preliminary web app v1, pilot backtest.
Nov 1–Nov 15	User Study R1	Recruit 15 participants, execute double-blind evaluation, log Likert scores.
Nov 16–Nov 22	Backtest Exec	Full 2015–2024 temporal backtest across all 4 baselines; citation impact test.
Nov 23–Nov 29	Thanksgiving	Statistical data analysis; user feedback synthesis; frontend polish.
Nov 30–Dec 4	System Polish	Edge case handling ($d = \infty$ paths); D3 interactive graph optimization.
Dec 5–Dec 9	Final Delivery	10–12 page ACM final research paper, 10-minute video walkthrough, code repo.

7 Division of Labor: Who Did What

In accordance with course project policy, each partner's technical contributions are detailed below:

- Evan Lin (elin26@illinois.edu): Led data harvesting (110,482 OpenAlex papers), Gray Book entity matching, co-authorship graph construction, Louvain community discovery, D3.js visualization, and offline backtest baseline scripting.
- Eric Liu (ericl8@illinois.edu): Formulated mathematical brokerage scoring (Eq. (2)), Burt constraint derivation, FastAPI backend service, Dijkstra path router, double-blind user study protocol, and survey instrumentation.
- Joint Work: Both team members jointly authored the system architecture, resolved the topological distance framework (Section 3.3), conducted system testing, and co-authored this mid-semester report.

8 Generative AI Usage Disclosure

In strict compliance with the CS 470 Generative AI policy, we disclose our use of AI technologies. We used Claude (Anthropic) as an interactive thinking partner to brainstorm baseline comparisons, assist with LaTeX formatting, and refine mathematical notation. Both human team members independently derived all mathematical formulas, executed all data processing scripts on the OpenAlex dataset, wrote and tested the underlying code, and critically reviewed and verified every factual claim, equation, and citation in this submission.

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